Technical Assessment

SDE Intern Technical Assessment_XIPL

Total: 100 marks | Expected effort: 3 days | All models: open-source, free-tier friendly

1. Overview

This assessment evaluates your ability to build a working AI-based virtual try-on pipeline: understanding garments and people with a vision-language model, preparing images through parsing and segmentation, running an open-source try-on model end to end, evaluating output quality automatically, and shipping a simple web demo.

You are not expected to train any model. Every task uses open-source models with publicly downloadable weights that run on a free Google Colab T4 GPU. We are assessing how you select models, read research repositories, chain pipelines together, handle constraints, and document your work.

2. Rules

- Use only open-source models with freely downloadable weights. Paid or hosted APIs are not allowed for any model inference.
- Free-tier compute (Google Colab, Kaggle, Hugging Face Spaces) is permitted and expected.
- You may use AI coding assistants, but you must be able to explain every line of your submission in a follow-up interview.
- Try-on model weights (IDM-VTON, OOTDiffusion, CatVTON) carry non-commercial research licenses. That is acceptable for this assignment; note the license of every model you use in your README.
- Partial credit is given for well-documented failures. A clear explanation of what broke and what you tried is worth more than silent, broken code.

3. Provided sample files

The accompanying **sample_files.zip** contains everything referenced in the questions:

File / folder	Purpose
test_pairs/person/ (5 images)	Person photos, 768x1024. person_01 to person_03 form the official test pairs; person_04 and person_05 are additional inputs for Q1 and Q2.
test_pairs/garment/ (5 images)	Flat garment product photos, 768x1024, matched by number to the person images.
test_pairs/pairs_manifest.csv	Defines the official pairs (pair_01 to pair_03), flags which pairs contain built-in edge cases, and lists the two slots (pair_04, pair_05) you must fill with your own sourced images.
edge_cases/ (4 images)	Dedicated edge-case inputs: person_crossed_arms.jpg, person_side_pose.jpg, person_seated.jpg, and no_person.jpg. Used to test Q2 robustness and Q5 guardrails.

edge_cases/edge_cases_manifest.csv	Maps each edge-case image to the question it tests and the expected behavior of your pipeline.
sample_output_q1.json	Example of the exact JSON structure expected from your Q1 script.
evaluation_template_q4.csv	The comparison table you must fill with your Q4 scores and commit to your repo.
SUBMISSION_TEMPLATE.md	Copy this into your repo root as README.md and fill in every section.

For Q1, run your script on all 10 provided images (5 person + 5 garment). For Q3 to Q5, use the three official pairs plus two person/garment pairs you source yourself (royalty-free stock photos or your own photos; front-facing, upper body visible).

4. Question weightage summary

Question	Title	Weightage (marks)
Q1	Garment & body understanding with a vision-language model	15
Q2	Human parsing & garment segmentation pipeline	20
Q3	End-to-end try-on inference	25
Q4	Automated quality evaluation of try-on results	20
Q5	Mini try-on web demo	20
	Total	**100**

5. Questions

Q1. Garment & Body Understanding with a Vision-Language Model - 15 marks

Using one of these open VLMs - **LLaVA-NeXT (Apache 2.0), InternVL2 (MIT), MiniCPM-V 2.6 (Apache 2.0), or Florence-2 (MIT)** - build a script that takes a photo of a person and a photo of a garment and outputs structured JSON with: garment type, sleeve length, neckline, primary color and pattern, the person's pose category (front-facing / side / seated), and visibility of upper and lower body. Match the structure shown in sample_output_q1.json. Justify your model choice in the README.

Deliverables: script + JSON outputs for all 10 test_pairs images plus the 3 person images in edge_cases/ (the side and seated photos must be classified correctly). Estimated time: 0.5 day.

Q2. Human Parsing & Garment Segmentation Pipeline - 20 marks

Given a person image, produce (a) a human parsing map using an open model - **SCHP, Segment Anything + Grounding DINO, or DensePose** - and (b) an "agnostic" person representation with upper-body clothing masked out, the standard input format for diffusion-based try-on models. Also segment each garment cleanly from its product photo using an open background-removal model such as **RMBG-1.4 or rembg**. Pay

attention to edge cases, all of which are present in the provided files: hair over shoulders (person_02, person_03), a strappy sleeveless garment (garment_03), and crossed arms (edge_cases/person_crossed_arms.jpg). Include the parsing map and agnostic image for the crossed-arms photo in your outputs.

Deliverables: pipeline code + parsing maps, agnostic images, and garment masks for the provided images including the crossed-arms edge case. Estimated time: 0.75 day.

Q3. End-to-End Try-On Inference - 25 marks

Run an open-source try-on model - **IDM-VTON, OOTDiffusion, or CatVTON** (all weights freely available on Hugging Face) - end to end on the 5 pairs defined in pairs_manifest.csv, wiring your Q2 pipeline in as the preprocessing stage. Document every GPU or memory constraint you hit and the workaround you applied (quantization, CPU offload, reduced resolution). CatVTON is the lightest and most Colab-friendly option if hardware is a constraint.

Deliverables: inference code + 5 try-on output images + constraints log in the README. Estimated time: 1 day.

Q4. Automated Quality Evaluation of Try-On Results - 20 marks

Build an evaluation script that scores each Q3 output on three axes: (a) **garment fidelity** - similarity between the input garment and the garment region of the output, using OpenCLIP or DINOv2 embeddings; (b) **identity preservation** - face-region SSIM or embedding distance using InsightFace; and (c) a **VLM-as-judge** score using your Q1 model with a written rubric prompt covering fit realism, artifacts, and texture transfer, returning a 1-10 score with reasons. Fill in evaluation_template_q4.csv for all 5 pairs and commit it to your repo.

Deliverables: evaluation script + completed CSV + your judge rubric prompt in the README. Estimated time: 0.5 day.

Q5. Mini Try-On Web Demo - 20 marks

Wrap Q2 to Q4 into a simple **Gradio or Streamlit** app: upload a person image and a garment image, and display the try-on result alongside the Q4 quality scores and the Q1 garment attributes. Implement guardrails: reject images where no person is detected, warn when the pose is unsuitable (seated or side view), and show an estimated processing time. Your demo video must show the guardrails firing on the three provided test inputs: edge_cases/no_person.jpg (rejected), edge_cases/person_seated.jpg (warning), and edge_cases/person_side_pose.jpg (warning).

Deliverables: app code + demo video showing guardrails triggered by the edge_cases images. Estimated time: 0.75 day.

6. Submission requirements

- A public or invite-shared GitHub repository with one folder per question and the SUBMISSION_TEMPLATE.md filled in as the root README.
- A demo video of at most 5 minutes walking through your working pipeline and web demo.
- Links to any Colab notebooks used, with outputs saved, so results can be verified quickly.

- The completed evaluation_template_q4.csv committed to the repo.

7. How you will be scored

Each question is scored as: **50%** working code that runs from the README instructions, **30%** output quality, and **20%** code quality and documented trade-offs. Well-explained failures receive partial credit; undocumented broken code receives none.

Good luck - we look forward to seeing your work.